Foundation for Advancement of Science & Technology

HW # 26

M. Abdullah Siddiqui 23I-0617 CS-F

(a) (consider v an eigenvector & λ an eigenvalue

i)

$$Ax = \lambda x$$

$$Av = \lambda v \rightarrow (i)$$

$$(A - cI)v = \lambda v$$

$$Av - cv = \lambda v$$

By (i)

$$\lambda v - cv = \lambda v$$

$$v(\lambda - c) = v$$

$\rightarrow$ v is an eigenvector

λ-c is eigenvalue

ii)

$$Ax = \lambda x$$

$$Av = \lambda v \rightarrow (i)$$

$$A^{-1} v = \lambda v$$

$$\frac{1}{\lambda} v = \lambda v \qquad \because A^{-1} v = \frac{1}{\lambda} v$$

iii)

$$Av = \lambda v$$

$$A^K v = \lambda v$$

$$\lambda^K v = \lambda v$$

λ^K is eigen value & v eigenvector

ii) Determinant of a matrix is the product of its eigenvalues. since det = 0. It is not invertible.

(b)

$P(\lambda) = (\lambda-2)(\lambda-3)$

$P(\lambda) = \lambda^2 - 5\lambda + 6$ characteristic polynomial

ii) Since matrices are similar, their eigenvalues will be same.

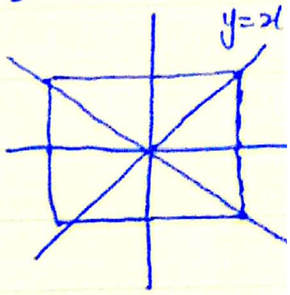
eigenvalues = 2 & 3.

Det $= 2 \times 3 \Rightarrow 6$

trace $= 2+3 \Rightarrow 5$

(c)

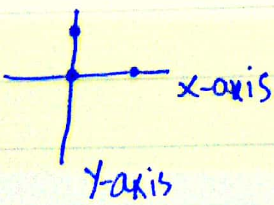
Transformation across $y=x$ remains unchanged in first quadrant. e.g, $\begin{bmatrix}1\\1\end{bmatrix}$ remains $\begin{bmatrix}1\\1\end{bmatrix}$ when multiplied by 1. $\lambda=1$ $v=\begin{bmatrix}1\\1\end{bmatrix}$



But $y=-x$, points change their signs, e.g, $\begin{bmatrix}1\\-1\end{bmatrix}$ changes to $\begin{bmatrix}-1\\1\end{bmatrix}$ when multiplied by -1. $\lambda=-1$ $v=\begin{bmatrix}-1\\1\end{bmatrix}$

ii) for projection on x-axis:

→ Points on x-axis remains unchanged. $\lambda=1$



→ Points on y-axis are projected on x-axis. $\lambda=0$

e.g: $\begin{bmatrix}1\\0\end{bmatrix} \to \begin{bmatrix}1\\0\end{bmatrix}$ $\begin{bmatrix}0\\1\end{bmatrix} \to \begin{bmatrix}0\\0\end{bmatrix}$

eigenspace eigenspace

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Contraction collapses vectors in y-direction to origin while in the x-direction to reduction in its magnitude.

e.g. $0 \le K < 1$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} K \\ 0 \end{bmatrix} \quad 0 < K < 1 \quad , \lambda = 1$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad K = 0 \quad , \lambda = 0$$

eigenspace